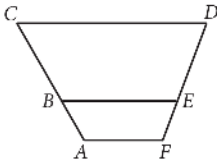
Question ID: 81b664bc

Question



In the figure above, $\overline{AF}$, $\overline{BE}$, and $\overline{CD}$ are parallel. Points B and E lie on $\overline{AC}$ and $\overline{FD}$, respectively. If $AB = 9$, $BC = 18.5$, and $FE = 8.5$, what is the length of $\overline{ED}$, to the nearest tenth?

Answer

A. 16.8

B. 17.5

C. 18.4

D. 19.6

Question ID: 94364a79

Question

Two nearby trees are perpendicular to the ground, which is flat. One of these trees is 10 feet tall and has a shadow that is 5 feet long. At the same time, the shadow of the other tree is 2 feet long. How tall, in feet, is the other tree?

Answer

A. 3

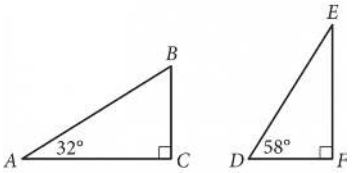
B. 4

C. 8

D. 27

Question ID: 933fee1a

Question



Triangles ABC and DEF are shown above. Which of the following is equal to the ratio $\frac{BC}{AB}$?

Answer

A. $\frac{DE}{DF}$

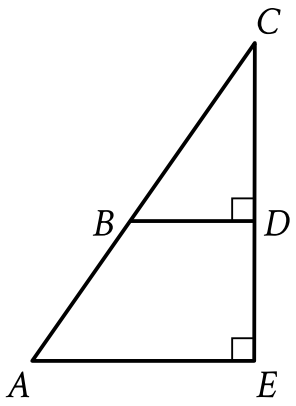
B. $\frac{DF}{DE}$

C. $\frac{DF}{EF}$

D. $\frac{EF}{DE}$

Question ID: 2f7c92ad

Question



Note: Figure not drawn to scale.

In the figure shown, triangle CAE is similar to triangle CBD. The measure of angle CBD is $57°$, and $AE = 26(BD)$. What is the measure of angle CAE?

Answer

A. $(26 \cdot 57)°$

B. $(26 + 57)°$

C. $57°$

D. $26°$

Question ID: d3fe472f

Question

Triangle ABC is similar to triangle XYZ, such that A, B, and C correspond to X, Y, and Z respectively. The length of each side of triangle XYZ is 2 times the length of its corresponding side in triangle ABC. The measure of side AB is 16. What is the measure of side XY?

Answer

A. 14

B. 16

C. 18

D. 32

Question ID: 2085e10e

Question

In triangle DEF, the measure of angle D is $47°$ and the measure of angle E is $97°$. In triangle RST, the measure of angle R is $47°$ and the measure of angle S is $97°$. Which of the following additional pieces of information is needed to determine whether triangle DEF is similar to triangle RST?

Answer

A. The measure of angle F

B. The measure of angle T

C. The measure of angle F and the measure of angle T

D. No additional information is needed.

Question ID: 901e3285

Question

In triangle ABC, the measure of angle A is $50°$. If triangle ABC is isosceles, which of the following is NOT a possible measure of angle B ?

Answer

A. $50°$

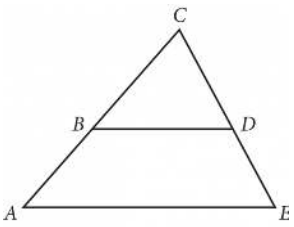
B. $65°$

C. $80°$

D. $100°$

Question ID: 6dd463ca

Question



Note: Figure not drawn to scale.

In the figure above, segments AE and BD are parallel. If angle BDC measures 58° and angle ACE measures 62°, what is the measure of angle CAE ?

Answer

A. 58°

B. 60°

C. 62°

D. 120°

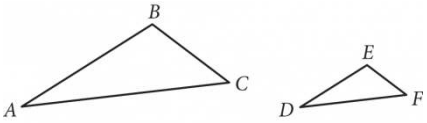
Question ID: 7a8ad237

Question

Triangles ABC and DEF are congruent, where A corresponds to D, and B and E are right angles. The measure of angle A is $69°$. What is the measure, in degrees, of angle F?

Question ID: 1c3d613c

Question

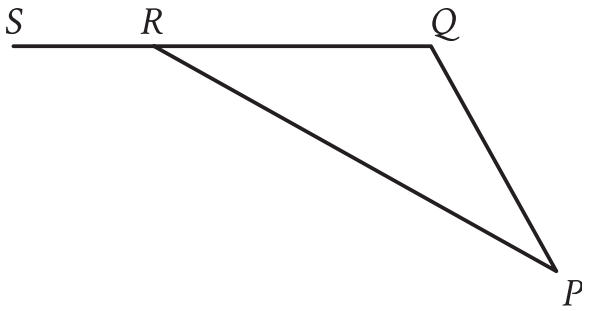


Note: Figures not drawn to scale.

Triangle ABC and triangle DEF are shown. The relationship between the side lengths of the two triangles is such that $\frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF} = 3$. If the measure of angle BAC is 20°, what is the measure, in degrees, of angle EDF ? (Disregard the degree symbol when gridding your answer.)

Question ID: 014edcb7

Question



Note: Figure not drawn to scale.

In triangle PQR, $\overline{QR}$ is extended to point S. The measure of $\angle PQR$ is $132°$, and the measure of $\angle PRS$ is $163°$. What is the measure of $\angle QPR$?

Answer

A. $48°$

B. $31°$

C. $24°$

D. $17°$